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19 December 1986
SUPERSEDING
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28 May 1974

MILITARY SPECIFICATION

# TRUCKS, HAND, LIFT; PALLET, MANUAL-HYDRAULIC

This specification is approved for use within the USA Belvoir Research, Development, and Engineering Center, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

## 1. SCOPE

1.1 Scope. This specification covers two sizes and four classes of general purpose, heavy-duty, manually operated, industrial-type hand lift pallet trucks.

1.2 Classification. Hand lift pallet trucks shall be of the following sizes and classes as specified (see 6.2):

Size A - 4,000-pound pallet load capacity.
Size B - 6,000-pound pallet load capacity.
Class 1 - 40-inch for length.
Class 2 - 48-inch fork length.
Class 3 - 36-inch fork length.
Class 4 - 42-inch fork length.

## 2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. The following specifications and standards form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DoDISS) and supplement thereto, cited in the solicitation.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: USA Belvoir Research, Development, and Engineering Center, ATTN: STRBE-TSE, Fort Belvoir, VA 22060-5606 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 3920

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SPECIFICATIONS

FEDERAL

| | |
|---|---|
| QQ-S-781 | - Strapping, Steel, and Seals. |
| PPP-B-601 | - Boxes, Wood, Cleated Plywood. |
| PPP-B-621 | - Boxes, Wood, Nailed and Lock-Corner. |
| PPP-T-97 | - Tape, Pressure Sensitive Adhesive, Filament Reinforced. |

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|---|---|
| MIL-P-116 | - Preservation, Methods of. |
| MIL-L-2105 | - Lubricating Oil, Gear, Multipurpose. |
| MIL-G-10924 | - Grease, Automotive and Artillery. |
| MIL-P-15011 | - Pallets, Material Handling, Wood, Post Construction, 4-Way Entry. |

STANDARDS

FEDERAL

| | |
|---|---|
| FED-STD-H28 | - Screw Thread Standards for Federal Services. |

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|---|---|
| MIL-STD-105 | - Sampling Procedures and Tables for Inspection by Attributes. |
| MIL-STD-129 | - Marking for Shipment and Storage. |
| MIL-STD-889 | - Dissimilar Metals. |
| MIL-STD-1186 | - Cushioning, Anchoring, Bracing, Blocking and Waterproofing with Appropriate Test Methods. |

(Copies of specifications, standards, and other Government documents required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

2.1.1 Other Government documents. The following other Government documents form a part of this specification to the extent specified herein. Unless otherwise specified, the issues shall be those in effect on the date of the solicitation.

DEPARTMENT OF LABOR (DOL)

29 CFR 1910 - Occupational Safety and Health Administration (OSHA)

(Application for copies should be addressed to a DOL officer or the Superintendent of Documents, US Government Printing Office, Washington, DC 20402.)



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2.2 Other publications. The following document(s) form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted shall be those listed in the issue of the DoDISS specified in the solicitation. Unless otherwise specified, the issues of documents not listed in the DoDISS shall be the issue of the nongovernment documents which is current on the date of the solicitation.

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

D 3951 - Standard Practice for Commercial Packaging.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

SOCIETY OF AUTOMOTIVE ENGINEERS (SAE)

SAE Handbook

(Application for copies should be addressed to the Society of Automotive Engineers, 400 Commonwealth Drive, Warrendale, PA 15086.)

(Non-Government standards and other publications are normally available from the organizations which prepare or which distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Description. The hand lift pallet trucks (hereinafter called "trucks") shall be heavy-duty industrial-type. The rated load-carrying capacities of the trucks shall be 4,000 and 6,000 pounds. Components shall consist of the hand propelled and hand operated hydraulic lift, multiple stroke; polyurethane tires cured-on dual front wheels, and rear wheels with cured-on polyurethane tires. The length of the fork shall be 36, 40, 42, or 48 inches as specified (see 1.2).

3.2 Standard product. Except as otherwise specified herein, the trucks shall be the standard product of the contractor and shall comply with all federal laws including DOL-OSHA standards and regulations. The trucks shall be new and unused except to the extent required to perform the tests specified herein.

3.3 First article. When specified (see 6.2), a sample shall be subjected to first article inspection (see 4.3 and 6.3). Any changes or deviations of trucks from the approved first article during production will be subject to the approval of the contracting officer. Approval of the first article will not relieve the contractor of his obligation to furnish trucks conforming to this specification.



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3.4 Material. Material shall be as specified herein. Materials not specified shall be selected by the contractor and shall be subject to all provisions of this specification.

3.4.1 Material deterioration prevention and control. The hand lift trucks shall be fabricated from compatible materials, inherently corrosion resistant or treated to provide protection against the various forms of corrosion and deterioration that may be encountered in any of the applicable operation and storage environment to which the hand lift trucks may be exposed.

3.4.2 Dissimilar metals. Dissimilar metals shall not be used in intimate contact with each other unless protected against galvanic corrosion. Dissimilar metals and methods of protection are defined and detailed in MIL-STD-889.

3.4.3 Identification of materials and finishes. The contractor shall identify the specific material, material finish or treatment for use with component and subcomponent, and shall make information available upon request to the contracting officer or designated representative.

3.4.4 Recovered materials. For the purpose of this requirement, recovered materials are those materials which have been collected from solid waste and reprocessed to become a source of raw materials, as distinguished from virgin raw materials. The components, pieces and parts incorporated in the hand lift trucks may be newly fabricated from recovered materials to the maximum extent practicable, provided the hand lift trucks produced meet all other requirements of this specification. Used, rebuilt or remanufactured components, pieces and parts shall not be incorporated in the hand lift trucks.

3.5 Construction.

3.5.1 Chassis and frame. The frame and related structure shall provide a rigid unit. The overall length of the trucks, except class 3 with handle in the vertical position, shall not exceed a distance equal to the fork length plus 24 inches. For class 3 trucks the overall length shall not exceed the fork length plus 18 inches.

3.5.2 Forks. Forks shall be fabricated of formed steel channel cross sections. Minimum under-clearance, with forks lowered, shall be 1 inch. At any elevation, forks shall be horizontal within a tolerance of plus 1/16 inch and minus 1/8 inch at the ends and in alinement within a tolerance of plus or minus 1/4 inch. Forks shall be end tapered on the underside and sides to facilitate entry and withdrawal from standard military pallets conforming to MIL-P-15011. The lowered fork height shall not exceed 3-1/4 inches. The overall dimensions for forks shall be as specified in table I.



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TABLE I. Dimensions.

| Truck class | Fork length Min. (inches) | Fork length Max (inches) | Spacing between forks (Min.) (inches) | Distance between outer edges of forks (Max.) (inches) |
|---|---|---|---|---|
| 1 | 39-1/2 | 40 | 8-3/4 | 27 |
| 2 | 47-1/2 | 48 | 8-3/4 | 27 |
| 3 | 35-1/2 | 36 | 8 | 22 |
| 4 | 41-1/2 | 42 | 8-3/4 | 27 |

3.5.3 Raising and lowering mechanism. When using a pallet conforming to MIL-P-15011, for class 1 truck, and with the truck fully inserted into the pallet at a right angle to the bottom deckboard, the raising and lowering mechanism shall permit the rear wheels to be lowered and raised in the bottom deckboard openings with a minimum clearance of 1/4 inch between the bottom deckboards and the wheels. The mechanism shall be recessed, or otherwise protected, to prevent damage from impact and abrasion during normal operation. Linkages and wheel supports shall move freely without dragging on pallet boards during raising and lowering operation. The mechanism shall be capable of returning unloaded rear wheels into the recessed position. All other classes shall be inserted into the pallet parallel to the bottom deckboard.

3.5.4 Wheels and rollers.

3.5.4.1 Front wheels. Unless otherwise specified (see 6.2), each truck shall be equipped with dual front wheels. Wheels shall have cured-on, polyurethane tires and ball or roller bearings. Wheels shall be steel, aluminum, or malleable or cast iron and shall be equipped with permanent lifetime grease-sealed bearings or accessible pressure lubrication fittings. Each wheel of dual front wheels shall be not less than 8 inches in diameter by not less than 2 inches wide.

3.5.4.2 Rear wheels. Unless otherwise specified (see 6.2), rear wheels shall be steel or aluminum with cured-on polyurethane tires. Rear wheels shall be concentric within .020 total indicated reading. Wheels shall be equipped with ball or roller bearings. Wheels shall be provided with permanent lifetime grease-sealed bearings or accessible pressure lubrication fittings. Each wheel shall be not less than 3 inches in diameter by not less than 3-3/4 inches in width.

3.5.4.3 Entry and withdrawal devices. Pallet entry and withdrawal devices shall be provided, and shall be in accordance with the contractor's standard practice except as specified herein. The entry and withdrawal devices shall have an underclearance of not less than 1/8 inch.



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3.5.5 Steering. The trucks shall be steered through the dual front wheels by a steering handle. The steering handle shall be fabricated of steel, and shall be capable of turning the steering assembly not less than 100 degrees to the left and 100 degrees to the right of dead center. The steering assembly shall be mounted on an angular contact ball or tapered roller bearings. The steering assembly shall have permanent lifetime grease-sealed bearings, or provisions for lubrication by accessible pressure fittings. The steering handle shall be of the balanced type and capable of remaining in a vertical position. The handle shall not have a snap action back to the vertical position when extended to any position. When, the handle is extended to 30 to 42 inches above ground level it may "creep" either upward or downwards. The handle shall be capable of moving through a vertical arc of not less than 65 degrees measured from the handle's normal upright position. The handle height shall be not more than 32 inches above ground level after swinging through the full vertical arc. The handle shall be 5 degrees to 10 degrees to the rear of the vertical when in the released position. The height of the steering handle shall be not more than 51 inches when measured vertically from floor level. The steering handle shall be removable from the steering mechanism with the use of common handtools.

3.5.6 Hydraulic system. The hydraulic system of the trucks, including the pump, ram, and reservoir, shall be the sealed type unit(s). The sealed type unit(s) shall be demountable and shall hold a rated load in any position without exceeding a downward drift rate of 1/8 inch per hour when tested as specified in 4.5.2.3. Provision shall be made to control lowering speed to prevent jarring. Component mechanisms used in raising and lowering shall be protected from damage. The hydraulic reservoir shall be provided with an accessible filler plug for filling and checking the fluid level, and shall have sufficient capacity to fill the hydraulic system plus 30 percent reserve fluid. The hydraulic fluid fill port shall be labeled "HYDRAULIC FLUID". The system shall conform to MIL-L-2105. All hydraulic system component material shall be compatible with these fluids. A totally enclosed pressure-relief device shall be provided which shall automatically preclude any damage to the hydraulic system due to overload and continuous operation of the lift mechanism after the load is lifted to maximum height.

3.5.6.1 Ram. A wiper shall be furnished to prevent grit from entering the hydraulic system. The ram shall be chrome plated, in accordance with contractor's standard commercial practices to prevent rust while in an unprotected condition against the environment or while in storage.

3.5.6.2 Lifting and lowering mechanism. The lifting mechanism shall be controlled and operated by either the handle of the truck, a lever separate and independent of the handle, or by a foot pedal. The lowering mechanism shall be a pressure-release device that is independent of the lifting mechanism. The mechanism shall be so located that the operator will not be endangered when lowering a load. All hand-release devices shall withstand a force of 100 pounds and all foot pedal devices shall withstand 250 pounds without damage or malfunction.

3.6 Performance.



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3.6.1 Lifting operation. The lift mechanism shall be capable of raising and lowering the forks with the rated capacity load. The number of strokes and the force required to lift the rated load 4 inches above the lowered position shall be as specified in table II, when tested as specified in 4.5.2.2.

TABLE II. Lifting requirements.

| | Size A truck | | Size B truck | |
|---|---|---|---|---|
| Lifting mechanism | Maximum No. of strokes | Maximum force in pounds | Maximum No. of strokes | Maximum force in pounds |
| Handle | 14 | 45 | 20 | 45 |
| Lever | 25 | 30 | 38 | 30 |
| Foot Pedal | 24 | 100 | 34 | 100 |

3.6.2 Maximum load. The truck frame, forks, and related structure shall be capable of withstanding 200 percent of rated load, with forks elevated to maximum lift height, with no evidence of damage or permanent deformation exceeding 1/16 inch, when tested as specified in 4.5.2.3.

3.6.3 Shock load. Trucks equipped with a preset pressure-relief valve shall be loaded with 125 percent of the rated load. Trucks not equipped with a preset pressure-relief valve shall be loaded with 150 percent of the rated load. The truck shall show no evidence of settling, damage, permanent deformation, or distortion or binding of the wheel bearings, when tested as specified in 4.5.2.4.

3.6.4 Pallet entry and withdrawal. The truck shall be capable of complete entry into and withdrawal from an empty double faced, 4-way standard military pallet conforming to MIL-P-15011, without moving the pallet more than 6 inches in either direction, when tested as specified in 4.5.2.5.

3.6.5 Steering. The truck with rated load and the handle in the towing position shall not require a force in excess of 10 pounds for size A trucks and 15 pounds for size B trucks to turn the steering assembly, when tested as specified in 4.5.2.6.

3.6.5.1 Steer handle. The handle shall withstand a minimum force of 200 pounds applied horizontally to the handle, with handle being pulled toward the forks from the handle's normal upright position without damage or permanent deformation, when tested as specified in 4.5.2.6.1.

3.6.6 Hydraulic system. The hydraulic system shall show no evidence of a leakage or permanent deformation, when tested as specified in 4.5.2.3, 4.5.2.4, and 4.5.2.7.



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3.6.7 Starting force. The force required to start moving the truck shall be not more than 3 percent of the combined weight of the load and the truck, when tested as specified in 4.5.2.8.

3.7 Bearings. Bearings shall be sealed or shielded to retain grease and exclude dust and moisture. Tapered roller bearings shall provide a means of adjustment to compensate for wear.

3.8 Lubrication. Lubrication means shall be provided for all moving parts that require lubrication, except where sealed permenant lubrication is provided. Each truck shall be serviced with lubricants conforming to MIL-L-2105 and MIL-G-10924. Oil drain plugs and connections shall provide for drainage into a container using common maintenance shop tools. All seals and packing shall be compatible with the above specified lubricants.

3.8.1 Grease fittings. Components requiring grease lubrication shall be provided with pressure grease fittings in accordance with SAE J534.

3.9 Finish.

3.9.1 Surface preparation. All exterior surfaces shall be cleaned to provide a surface free from mill scale, oil, grease, dirt, and rust in accordance with the contractor's standard practice.

3.9.2 Painting. The cleaned and dried surfaces prepared in accordance with 3.9.1 shall be painted with not less than one coat of metallic base primer and two coats of synthetic enamel. If an airless paint spray process is used, only one finish is required. The primer and finish coats shall be supplied in accordance with the contractor's standard practice. Unless otherwise specified (see 6.2), the finish coat shall be bright yellow or any bright safety color. The finish coat when dry shall have a smooth, even surface, free from runs, sags, and foreign material. Surfaces of components and assemblies not normally painted with a finish coat, shall be cleaned and prepared in accordance with the contractor's standard practice.

3.10 Identification plate. The contractor shall provide, and permanently attach by screws, bolts, rivets, or spot welds, a corrosion-resistant metal identification plate. The plate shall be installed in a protected location where the plate shall be both visible and legible on the truck with the following information indented, embossed, or otherwise permanently marked on the plate:

TRUCK, HAND, LIFT: PALLET, MANUAL-HYDRAULIC
CAPACITY
NAME OF MANUFACTURER
SERIAL NUMBER
MODEL NUMBER
DELIVERY DATA (month and year)
CONTRACT OR ORDER NUMBER
FEDERAL STOCK NUMBER
US



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3.11 Maintenance. Provisions shall be made for adjustment, service, and replacement of wheels, axles, wearing parts of lift mechanism, steering assembly, and hydraulic pumps. Any frequently adjusted or replaced items shall be easily accessible. There shall be no interference to the servicing and draining of oils and lubricants to or from any assembly or components by frame members or other obstruction.

3.12 Equipment manual. Unless otherwise specified (see 6.2), the contractor shall furnish one copy of an equipment manual with each truck. The equipment manual shall contain complete instructions for operation and maintenance, and illustrated repair parts section. The repair parts section shall include actual manufacturer's name and part number and quantity of each part required. The repair part illustration may be a drawing or picture. Repair parts for the same or different model trucks shall not have the same part number unless the parts are interchangeable.

3.13 Workmanship. The trucks shall withstand any operation specified herein without permanent deformation, breakage of connections, malfunction, or component interference caused by incorrect workmanship. All parts of the trucks before and after painting shall be clean and free from sand, rust, dirt, fins, sprues, scale, flux, and other harmful extraneous materials. Edges and surfaces exposed to operating and maintenance personnel shall be smooth and rounded to the extent that a hazardous surface does not exist.

3.13.1 Castings. Castings shall be of uniform quality and condition, free from blowholes, porosity, hard spots, shrinkage defects, and cracks. Castings shall be commercially machineable and shall not be repaired, plugged or welded.

3.13.2 Bolts and nuts. Boltholes shall be accurately formed. Washers or lockwashers or a combination of the two shall be provided on all bolts, studs, and capscrews having straight threads. Self-locking nuts are acceptable in lieu of standard nuts and lockwashers. All fasteners shall be correctly torqued and nuts shall have full thread engagement. Bolts shall not extend through nuts more than four full threads.

3.13.2.1 Screw threads. Screw threads shall be in accordance with FED-STD-H28.

3.13.3 Welds. Welded joints shall be sound, smooth, free from holes, fissures, rough projecting edges, and slag, and shall be thoroughly fused to the base metal.

## 4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the



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right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.2 Classification of inspections. Inspections shall be classified as follows:

a. First article inspection (see 4.3).
b. Quality conformance inspection (see 4.4).
c. Inspection of packaging (4.6)
d. Certification of Conformance (4.7).

4.3 First article inspection.

4.3.1 Examination. The truck shall be examined as specified in 4.5.1. Presence of one more defects shall be cause for rejection.

4.3.2 Tests. The truck shall be tested as specified in 4.5.2.1 through 4.5.2.8. Failure of any test shall be cause for rejection.

4.4 Quality conformance inspection.

4.4.1 Sampling. Sampling for examination and tests shall be in accordance with MIL-STD-105. A lot shall consist of all trucks offered for acceptance at one time. The sample unit for this inspection shall be one fully assembled truck.

4.4.2 Examination. Samples selected in accordance with 4.4.1 shall be examined as specified in 4.5.1. AQL shall be 1.0 percent defective for major defects and 2.0 percent for minor defects.

4.4.3 Test. Samples selected in accordance with 4.4.1 shall be tested as specified in 4.5.2.1, 4.5.2.2, and 4.5.2.7. AQL shall be 1.0 percent defective.

4.5 Inspection procedure.

4.5.1 Examination. The trucks shall be examined as specified herein for the following defects:

Major

101. Dimensions not as specified.
102. Any characteristic not in accordance with specified requirements.
103. Part missing or not specified type or size.



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104. Materials not as specified.
105. Materials not resistant to corrosion and deterioration, or treated to be resistant to corrosion and deterioration for the applicable storage and operating environments.
106. Dissimilar metals as defined in MIL-STD-889 are not effectively insulted from each other.
107. Contractor does not have documentation available for identification of material, material finishes, or treatment.
108. Used, rebuilt or remanufactured components, pieces, or parts incorporated in the hand lift truck.
109. Assembly or component fractured, split, bowed, or malformed.
110. Part misplaced, loose, or not in correct alinement.
111. Sharp burr, sliver, or splinter.
112. Castings not as specified.
113. Welds not as specified, missing, incomplete, cracked, not correctly fused.
114. Lubrication and fittings not as specified.
115. Forks not alined as specified.
116. Evidence of rust.
117. Treatment and painting not as specified.
118. Bolts and nuts not as specified.
119. Screw threads not as specified.
120. Bearings not as specified.
121. Lubrication not as specified.
122. Front and rear wheels not as specified.

Minor

201. Identification marking missing, incomplete, not legible.
202. Equipment manual missing, incomplete, not legible.
203. Threaded fasteners missing, broken, stripped or loose.
204. Finish not dry, free of runs, sags, or area of no film.

4.5.2 Test.

4.5.2.1 *Test conditions*. Prior to test, the truck shall be lubricated with oils and greases as recommended by the contractor. All tests unless otherwise



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specifed herein, shall be conducted using the standard truck which has been supplied. Test shall be conducted in the order listed.

4.5.2.2 Lifting operation. With the rated load on the forks, record the number of strokes and force required to lift the rated load four inches above the lowered position. Nonconformance to 3.6.1 shall constitute failure of this test.

4.5.2.3 Maximum load. Prior to conducting this test, and the test specified in 4.5.2.4, the pressure-relief valve shall be made inoperative, if applicable, and the truck shall be equipped with steel wheels. With the truck placed on a surface plate and with forks at maximum height, the following heights shall be measured, the points of measurement marked on the truck, and recorded:

a. Front end of frame.
b. Front pivot point of raising and lowering mechanism.
c. Top of forks at heel on both sides.
d. Top of forks midway between front and rear pivot points of raising and lowering mechanism on both sides.
e. Rear pivot point of raising and lowering mechanism on both sides.

Then place the truck on a smooth, dry, level concrete surface with the forks maintained at maximum height. Place on the elevated forks a load of 8,000 pounds for a size A truck and a load of 12,000 pounds for a size B truck, as applicable. Measure the height of ram and at the end of 1 hour, measure the downward drift; remove the load and place the truck on a surface plate. Repeat measurements a. through e. Examine the truck for evidence of downward drift, permanent deformation, leakage, damage, and settling. Downward drift exceeding 1/8 inch, any variation in height dimensions in excess of 1/16 inch, or evidence of damage or permanent deformation in excess of 1/16 inch shall constitute failure of this test.

4.5.2.4 Shock load. Place an empty truck on a 1-inch steel plate platform and distribute uniformly the load specified in 3.6.3. Roll the truck off the steel platform onto a concrete surface. Remove the load and repeat the complete cycle for a total of two additional times. Then remove the load and tighten the joints to compensate for settling. Place the truck, with forks at maximum height, on a surface plate and measure and record the height specified in 4.5.2.3, a. through e. Repeat the complete shock cycle as described above for an additional five times. Remeasure and record the heights as specified above. Examine the truck for damage, leakage, and permanent deformation. Examine the wheel bearings for distortion and binding. Any variation in measurements between the two series of measurements in excess of 1/16 inch, evidence of damage, leakage or permanent deformation in excess of 1/16 inch, or evidence of distortion or binding of the wheel bearings shall constitute failure of this test.

4.5.2.5 Pallet entry and withdrawal. An empty double faced, 4-way standard military pallet conforming to MIL-P-15011 shall be placed with its front edge on a reference line. Push the truck until its entry device contacts the chamfered surface on the front bottom deckboard of the pallet; push the truck into the pallet until the rear wheels have cleared the center bottom deckboard; measure



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the distance from the refernce line to the same reference points on the pallet edge. This distance should not exceed 6 inches. Remove the truck from the pallet. Reset the pallet on the refernce line and repeat this procedure four times. Chock the pallet in the rear to prevent entry movement, and draw a new reference line. Engage the truck into the pallet as stated above. Withdraw the truck from the pallet. Measure the maximum distance from the reference line to the same reference point on the pallet edge. This distance should not exceed 6 inches. Reset the pallet on the reference line and repeat this procedure four times. A measured distance in excess of 6 inches during any entry or removal shall constitute failure of this test.

4.5.2.6 Steering. Place the truck carrying a rated load, on a smooth, level concrete surface. Place the handle into a pull position and attach a tension scale. Record the force necessary to turn the handle simulating steering. A force exceeding 10 pounds for size A trucks and 15 pounds for size B trucks shall constitute failure of this test.

4.5.2.6.1 Steer handle. Place the truck on a level surface with the handle in the released upright position. Block and secure the truck to the floor. Attach a tension scale to the handle and apply a horizontal force of 200 pounds to the top of the handle, pulling toward the load end of truck. Nonconformance to 3.6.5.1 shall constitute failure of this test.

4.5.2.7 Leakage. After successful completion of the above tests, turn the truck on its side on a clean paper surface. Allow the truck to remain in this position for 5 minutes. Examine the actuator reservoir, the hydraulic lines, and the paper surface for oil and grease spots. Evidence of any leakage shall constitute failure of this test.

4.5.2.8 Starting force. Place the truck with its rated load on a smooth level concrete surface. Attach a tension scale, and with a slow steady pull determine the maximum reading on the scale necessary to start moving the truck. Non-conformance to 3.6.7 shall constitute failure of this test.

4.6 Inspection of packaging.

4.6.1 Quality conformance inspection of pack.

4.6.1.1 Unit of product. For the purpose of inspection, a completed pack prepared for shipment shall be considered a unit of product.

4.6.1.2 Sampling. Sampling for examination shall be in accordance with MIL-STD-105.

4.6.1.3 Examination. Samples selected in accordance with 4.6.1.2 shall be examined for the following defects. AQL shall be 1.0 percent defective.

| | | |
|---|---|---|
| 123. | Disassembly not as specified for level A or commercial. | |
| 124. | Removed components not secured as specified for level A or commercial. | |
| 125. | Lubrication not applied as specified for level A or commercial. | |
| 126. | Unpainted surfaces not coated with preservative as specified for level A. | |



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127. Hydraulic system not filled as specified for level A.
128. Manual not preserved and secured as specified for level A.
129. Shipping container not as specified for level A or commercial.
130. Blocking, bracing, and anchoring not adequate to prevent movement and damage as specified for level A.
131. Strapping not as specified for level A.

4.7 Certificate of Conformance. Contractor shall provide a Certificate of Conformance certifying that the truck complies with all applicable DOL-OSHA standards and regulations.

5. PACKAGING

5.1 Preservation. Preservation shall be level A or commercial as specified (see 6.2).

5.1.1 Level A.

5.1.1.1 Disassembly. The steering handle shall be removed. Any additional dissassembly shall be the minimum necessary to accomplish reduction in cubage in accordance with the contractor's recommendation. Removed bolts, nuts, and washers shall be reinstalled in one of the mating parts and secured to prevent loss. The steering handle and any other removed components shall be secured to the basic unit in a protected location with soft annealed wire or tape conforming to PPP-T-97, type and size optional, and in a manner so as not to increase cubage.

5.1.1.2 Lubrication. All parts having grease fittings shall be lubricated as recommended by the equipment manual or by the contractor.

5.1.1.3 Unprotected surfaces. Unpainted exterior metal surfaces of the truck requiring the application of a contact preservative shall be coated with type P-1 preservative. The preservative shall conform to the applicable specification listed in and shall be applied in accordance with MIL-P-116.

5.1.1.4 Hydraulic system. The hydraulic system fluid supply tank shall be filled with the fluid recommended for operation. The piston shall be fully retracted into the cylinder and any exposed portions of the piston rod shall be coated with type P-1 preservative.

5.1.1.5 Equipment manual. The equipment manual shall be preserved in accordance with MIL-P-116, method IC-3, and secured to the truck with tape conforming to PPP-T-97, type and size optional, in a manner to prevent loss.

5.1.2 Commercial. Each truck shall be preserved in accordance with ASTM D 3951. Specifically, the steering handle shall be removed and secured to the truck to reduce cube to the minimum; or, if so designed, folded to its least protruding configuration. The truck shall be lubricated as recommended by the contractor.

5.2 Packing. Packing shall be level A or commerical as specified (see 6.2).



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5.2.1 Level A. Each complete truck, preserved as specified in 5.1, shall be packed in a box conforming to PPP-B-621, class 2, style 2; or to PPP-B-601, overseas type, style I. The truck shall be blocked, braced, and anchored within the box in accordance with MIL-STD-1186 to prevent movement or damage. The box shall be reinforced with strapping conforming to QQ-S-781, class 1, type I or IV, finish B.

5.2.2 Commerical. Each preserved truck shall be packed in accordance with ASTM D 3951 in a container which will allow efficient handling and stacking during shipment and storage.

5.3 Marking.

5.3.1 Level A. Marking shall be in accordance with MIL-STD-129.

5.3.2 Commercial. Marking shall be in accordance with ASTM D 3951. In addition, the gross weight and cube shall be marked on the shipping container.

6. NOTES

6.1 Intended use. Hand-lift pallet trucks described herein are intended for moving single or double faced pallet loads of material over short distances in and around warehouses, loading platforms, highway trucks, and railroad cars within the military supply system.

6.2 Ordering data. Procurement documents should specify the following:

a. Title, number, and date of this specification.
b. Size and class truck required (see 1.2).
c. Time frame required for submission of first article model and number of models required (see 3.3).
d. When the Government will conduct any or all of the first article model examination and tests. When the Government will conduct some but not all of the first article examination and tests, the contracting officer should specify which examination and test will be conducted by the Government and which examination and test shall be conducted by the contractor (see 3.3).
e. When front wheels other than dual are required (see 3.5.4.1).
f. When rear wheels other than steel or aluminum with cured-on polyurethane tires are required (see 3.5.4.2).
g. Color of finish coat if other than as specified (see 3.9.2).
h. When additional copies of equipment manual are required (see 3.12).
i. Degree of preservation and of packing required (see 5.1 and 5.2)
j. When other than finish B strapping is required (see 5.2.1).



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6.3 First article. When a first article inspection is required, the items should be a first produced model. The first article should consist of one unit. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examinations, tests, and approval of the first article test results and disposition of the first article.

6.4 Subject term (key word) listing.

a. Hand truck
b. Materials handling
c. Pallet
d. Truck

6.5 Changes from previous issue. Asterisks (or vertical lines) are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Custodians:
Army - ME

Preparing activity:
Army - ME

Project No. 3920-A162



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# STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

*(See Instructions – Reverse Side)*

| 1. DOCUMENT NUMBER | 2. DOCUMENT TITLE |
|---|---|
| MIL-T-52809A(ME) | Trucks, Hand, Lift; Pallet, Manual Hydraulic |

| 3a. NAME OF SUBMITTING ORGANIZATION | 4. TYPE OF ORGANIZATION *(Mark one)* |
|---|---|
| | ☐ VENDOR |
| b. ADDRESS *(Street, City, State, ZIP Code)* | ☐ USER |
| | ☐ MANUFACTURER |
| | ☐ OTHER *(Specify)*: _____ |

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

| 7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional | b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional |
|---|---|
| c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional | 8. DATE OF SUBMISSION *(YYMMDD)* |

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